$\neg r$: n is not even, i.e. n is odd (Defn. 1)

$\therefore$ m is odd and n is odd. From Defn. 1, $\exists$ integers K_1 and K_2 such that $m = 2K_1+1$ and $n = 2K_2+1$

$\therefore mn = (2K_1+1)(2K_2+1) = 4K_1K_2+2K_1+2K_2+1 = \boxed{2(2K_1K_2+K_1+K_2)+1}$

It's easy to see that $(2K_1K_2+K_1+K_2)$ is an integer. By Defn. 1, mn is odd

$\boxed{\therefore \neg p \text{ is True}}$. By defn. of conditional, $\boxed{\neg(q \vee r) \rightarrow \neg p}$ is True.

$\therefore$ $\boxed{p \rightarrow (q \vee r)}$ is True. By modus ponens, $(q \vee r)$ is True.

$\boxed{\text{Case 1:}}$ q is True, i.e. m is even.

We cover the checkerboard using the following strategy:

$\boxed{\text{Column 1:}}$ Row 1,2 $\rightarrow$ Dominoe 1 Row 3,4 $\rightarrow$ Dominoe 2, Row 5,6 $\rightarrow$ Dominoe 3, ..., Row $i, i+1$ $\rightarrow$ Dominoe $\left(\frac{i+1}{2}\right)$, ..., Row $m-1, m$ $\rightarrow$ Dominoe $m/2$

$\boxed{\text{Column 2:}}$ Row 1,2 $\rightarrow$ Dominoe $(m/2)+1$ Row 3,4 $\rightarrow$ Dominoe $(m/2)+2$, Row 5,6 $\rightarrow$ Dominoe $(m/2)+3$, ..., Row $i, i+1$ $\rightarrow$ Dominoe $m/2 + \frac{(i+1)}{2}$, ..., Row $m-1, m$ $\rightarrow$ Dominoe $m/2 + m/2$ = Dominoe $2m/2$

⋮

$\boxed{\text{Column j:}}$ Row 1,2 $\rightarrow$ Dominoe $(j-1)\left(\frac{m}{2}\right)+1$ Row 3,4 $\rightarrow$ Dominoe $(j-1)\left(\frac{m}{2}\right)+2$, Row 5,6 $\rightarrow$ Dominoe $(j-1)\left(\frac{m}{2}\right)+3$, ..., Row $i, i+1$ $\rightarrow$ Dominoe $(j-1)\left(\frac{m}{2}\right)+\frac{(i+1)}{2}$, ..., Row $m-1, m$ $\rightarrow$ Dominoe $(j-1)\frac{m}{2}+\frac{m}{2}$ = Dominoe $\frac{jm}{2}$

$\boxed{\text{Column n:}}$ Row 1,2 $\rightarrow$ Dominoe $(n-1)\left(\frac{m}{2}\right)+1$ Row 3,4 $\rightarrow$ Dominoe $(n-1)\left(\frac{m}{2}\right)+2$, ..., Row $i, i+1$ $\rightarrow$ Dominoe $(n-1)\left(\frac{m}{2}\right)+\frac{(i+1)}{2}$, ..., Row $m-1, m$ $\rightarrow$ Dominoe $(n-1)\frac{m}{2}+\frac{m}{2}$